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Fraud Detection — Industry Project

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Date: March 2025

Goal: Predict Transaction is fraud or not

==========

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

pd.set_option('display.max_columns',100)
pd.set_option('display.float_format', lambda x: '%.3f' % x)

print("All libraryies are successfully loaded !")
```

```
All libraryies are successfully loaded !
```

```
df = pd.read_csv('/content/creditcard.csv')
```

```
print('train_dataset shape :',df.shape)
```

```
train_dataset shape : (41683, 31)
```

```
print('Target columns (Class):')
print(df['Class'].value_counts())
```

```
Target columns (Class):
Class
0.000    41574
1.000      108
Name: count, dtype: int64
```

```
print(f"Fraud rate : {df['Class'].mean()*100:.2f}%")
```

```
Fraud rate : 0.26%
```

First look at data

```
df.head()
```

	Time	V1	V2	V3	V4	V5	V6	V7	V8
0	0	-1.360	-0.073	2.536	1.378	-0.338	0.462	0.240	0.099
1	0	1.192	0.266	0.166	0.448	0.060	-0.082	-0.079	0.085
2	1	-1.358	-1.340	1.773	0.380	-0.503	1.800	0.791	0.248
3	1	-0.966	-0.185	1.793	-0.863	-0.010	1.247	0.238	0.377
4	2	1.158	0.878	1.549	0.403	0.407	0.096	0.592	0.271

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 41683 entries, 0 to 41682
Data columns (total 31 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Time    41683 non-null  int64
 1   V1      41683 non-null  float64
 2   V2      41683 non-null  float64
 3   V3      41682 non-null  float64
 4   V4      41682 non-null  float64
 5   V5      41682 non-null  float64
 6   V6      41682 non-null  float64
 7   V7      41682 non-null  float64
 8   V8      41682 non-null  float64
 9   V9      41682 non-null  float64
 10  V10     41682 non-null  float64
 11  V11     41682 non-null  float64
 12  V12     41682 non-null  float64
 13  V13     41682 non-null  float64
 14  V14     41682 non-null  float64
 15  V15     41682 non-null  float64
 16  V16     41682 non-null  float64
 17  V17     41682 non-null  float64
 18  V18     41682 non-null  float64
 19  V19     41682 non-null  float64
 20  V20     41682 non-null  float64
 21  V21     41682 non-null  float64
 22  V22     41682 non-null  float64
 23  V23     41682 non-null  float64
 24  V24     41682 non-null  float64
 25  V25     41682 non-null  float64
 26  V26     41682 non-null  float64
 27  V27     41682 non-null  float64
 28  V28     41682 non-null  float64
 29  Amount  41682 non-null  float64
 30  Class   41682 non-null  float64
dtypes: float64(30), int64(1)
memory usage: 9.9 MB
```

```
df.describe()
```

	Time	V1	V2	V3	V4	
count	41683.000	41683.000	41683.000	41682.000	41682.000	416
mean	26208.346	-0.222	0.035	0.712	0.187	
std	12721.515	1.855	1.600	1.490	1.400	
min	0.000	-56.408	-72.716	-31.104	-5.173	-
25%	16853.000	-0.974	-0.533	0.230	-0.719	
50%	31305.000	-0.240	0.094	0.811	0.185	
75%	36289.000	1.159	0.740	1.443	1.069	

```
print(df.isnull().sum())
```

```
Time      0
V1        0
V2        0
V3        1
V4        1
V5        1
V6        1
V7        1
V8        1
V9        1
V10       1
V11       1
V12       1
V13       1
V14       1
V15       1
V16       1
V17       1
V18       1
V19       1
V20       1
V21       1
V22       1
V23       1
V24       1
V25       1
V26       1
V27       1
V28       1
Amount    1
Class     1
dtype: int64
```

```
missing_rows = df[df.isnull().any(axis=1)]
print("Missing rows : ",len(missing_rows))
print(missing_rows)
```

```
Missing rows :  1
        Time     V1     V2  V3  V4  V5  V6  V7  V8  V9  V10  V11  V1
41682  40784 -0.621 -0.734 NaN NaN NaN NaN NaN NaN NaN  NaN  NaN  Na

       V14  V15  V16  V17  V18  V19  V20  V21  V22  V23  V24  V25  V
41682  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN  NaN  N

       V28  Amount  Class
41682  NaN     NaN    NaN
```

```
df = df.dropna()
df.shape
```

```
(41682, 31)
```

```
df['Class'] = df['Class'].astype(int)
print("Class dtype : ", df['Class'].dtype)
```

```
Class dtype :  int64
```

```
print("Normal Transaction ")
print(df[df['Class']==0].describe())
```

```
Normal Transaction
           Time        V1        V2        V3        V4        V5
count 41574.000 41574.000 41574.000 41574.000 41574.000 41574.000 41
mean  26220.447    -0.204     0.021     0.741     0.172    -0.222
std   12723.790     1.758     1.558     1.287     1.364     1.320
min       0.000   -56.408   -72.716   -27.747    -5.173   -42.148
25%   16914.500    -0.969    -0.535     0.236    -0.721    -0.839
50%   31323.500    -0.236     0.091     0.815     0.180    -0.271
75%   36301.000     1.160     0.736     1.445     1.062     0.294
max   40783.000     1.960    14.846     4.102    16.491    34.802

             V7        V8        V9       V10       V11       V12
count 41574.000 41574.000 41574.000 41574.000 41574.000 41574.000 41
mean     -0.093     0.035     0.202    -0.050     0.398    -0.415
std       1.126     1.155     1.206     1.050     1.086     1.291
min     -26.548   -41.485    -4.533   -11.209    -4.050   -15.145
25%      -0.595    -0.151    -0.564    -0.545    -0.407    -1.220
50%      -0.072     0.050     0.074    -0.126     0.378    -0.093
75%       0.434     0.317     0.909     0.385     1.195     0.506
max      36.677    17.574    10.393    12.702    10.002     4.846

            V14       V15       V16       V17       V18       V19
count 41574.000 41574.000 41574.000 41574.000 41574.000 41574.000 41
mean      0.246     0.103     0.009     0.166    -0.084    -0.034
std       1.013     0.955     0.874     0.890     0.834     0.814
min     -13.780    -4.153   -10.116   -17.098    -5.367    -4.933
25%      -0.265    -0.457    -0.489    -0.345    -0.565    -0.525
50%       0.189     0.220     0.063     0.085    -0.073    -0.034
75%       0.782     0.782     0.558     0.572     0.403     0.474
max       7.692     5.785     6.099     9.254     4.296     4.555

            V21       V22       V23       V24       V25       V26
count 41574.000 41574.000 41574.000 41574.000 41574.000 41574.000 41
mean     -0.030    -0.111    -0.040     0.008     0.136     0.023
std       0.742     0.637     0.556     0.593     0.433     0.504
```

```
min     -20.262    -8.594   -26.751    -2.837    -7.496    -1.439
25%      -0.235    -0.532    -0.179    -0.324    -0.127    -0.330
50%      -0.075    -0.085    -0.051     0.061     0.176    -0.065
75%       0.101     0.301     0.077     0.401     0.421     0.303
max      22.615     5.806    17.298     4.014     5.525     3.517

            V28    Amount     Class
count 41574.000 41574.000 41574.000
mean      0.005    89.093     0.000
std       0.341   237.130     0.000
min      -9.618     0.000     0.000
25%      -0.007     7.550     0.000
50%       0.022    24.000     0.000
75%       0.076    80.000     0.000
max      33.848  7879.420     0.000
```

```
print("Fraud Transaction")
print(df[df['Class']==1].describe())
```

```
Fraud Transaction
           Time      V1      V2      V3      V4      V5      V6
count   108.000 108.000 108.000 108.000 108.000 108.000 108.000 108.
mean  21414.991  -7.455   5.595 -10.624   5.850  -5.116  -2.288  -7.
std   10822.969   9.305   4.826   9.583   2.858   6.994   1.630   7.
min     406.000 -30.552  -5.198 -31.104   0.817 -22.106  -4.978 -21.
25%   11121.250 -15.303   2.226 -19.955   3.771  -9.185  -3.358 -15.
50%   21232.500  -3.131   4.113  -5.923   6.028  -2.451  -2.328  -4.
75%   28947.750  -0.186   8.671  -3.226   6.669   0.030  -1.215  -1.
max   40742.000   1.234  16.713   2.041  11.928   5.269   3.628   5.

           V8      V9     V10     V11     V12     V13     V14     V1
count 108.000 108.000 108.000 108.000 108.000 108.000 108.000 108.00
mean    3.691  -2.905  -6.511   5.398  -8.130   0.346  -8.295  -0.00
std     6.140   2.264   4.126   3.071   4.516   0.926   4.457   0.79
min    -3.278  -7.175 -14.167  -0.415 -17.769  -2.648 -19.214  -2.08
25%     0.116  -4.080  -8.877   3.361 -10.565  -0.146 -10.721  -0.40
50%     1.015  -3.400  -6.573   4.899  -7.692   0.456  -7.866  -0.04
75%     3.912  -1.052  -3.530   6.834  -5.445   0.976  -5.329   0.50
max    20.007   2.821   3.245  12.019   1.238   1.734   1.501   2.00

          V17     V18     V19     V20     V21     V22     V23     V2
count 108.000 108.000 108.000 108.000 108.000 108.000 108.000 108.00
mean   -7.554  -2.667   0.428   0.631   0.603  -0.337  -0.314  -0.23
std     6.601   2.748   1.489   0.909   1.280   0.906   2.090   0.51
min   -18.587  -8.061  -3.286  -3.043  -2.476  -2.289 -19.254  -1.63
25%   -12.885  -4.883  -0.595   0.097   0.155  -0.890  -0.624  -0.56
50%    -9.867  -4.107   0.614   0.567   0.648  -0.273  -0.139  -0.08
75%    -2.771  -0.133   1.284   1.464   1.755   0.267   0.360   0.14
max     6.739   3.042   3.623   2.102   2.840   1.745   2.661   0.64

          V26     V27     V28   Amount   Class
count 108.000 108.000 108.000  108.000 108.000
mean    0.183   0.792   0.113   93.513   1.000
std     0.382   1.091   0.649  250.863   0.000
min    -0.657  -3.908  -1.869    0.000   1.000
25%    -0.210   0.263   0.047    1.000   1.000
50%     0.284   0.632   0.307    2.320   1.000
75%     0.524   1.574   0.488   99.990   1.000
max     0.817   3.052   1.109 1809.680   1.000
```

```
df['Hour'] = (df['Time'] % 3600) // 3600
```

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Exploratory Data Analysis (EDA)

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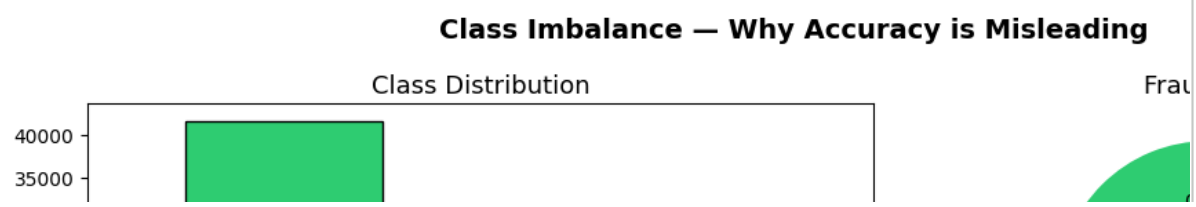
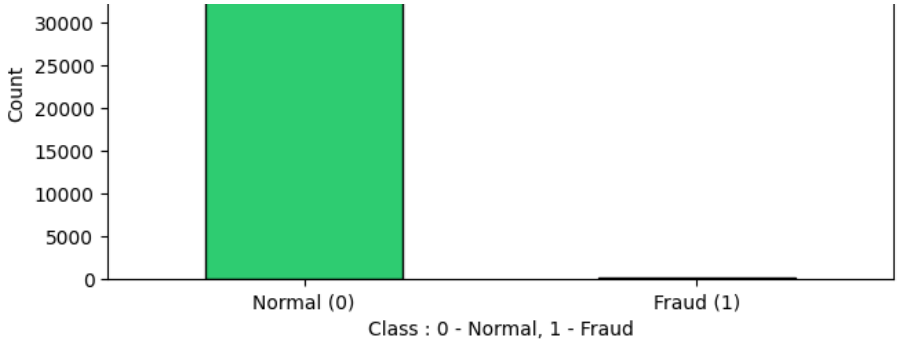
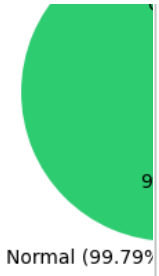
Goal: Keep Business insights

```
fig, axes = plt.subplots(1, 2 , figsize=(12,4))

df['Class'].value_counts().plot(
    kind='bar',
    ax = axes[0],
    color=['#2ecc71','#e74c3c'],
    edgecolor='black'
)
axes[0].set_title('Class Distribution',fontsize=13)
axes[0].set_xlabel('Class : 0 - Normal, 1 - Fraud')
axes[0].set_ylabel('Count')
axes[0].set_xticklabels(['Normal (0)', 'Fraud (1)'], rotation=0)

sizes = df['Class'].value_counts()
axes[1].pie(
    sizes,
    labels=['Normal (99.79%)', 'Fraud (0.21%)'],
    colors=['#2ecc71','#e74c3c'],
    autopct='%1.1f%%',
    startangle=90
)
axes[1].set_title('Fraud rate',fontsize=13)

plt.suptitle('Class Imbalance — Why Accuracy is Misleading',
             fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('01_class_distribution.png', dpi=150, bbox_inches='tigh
plt.show()

print("Bussiness Insights")
```

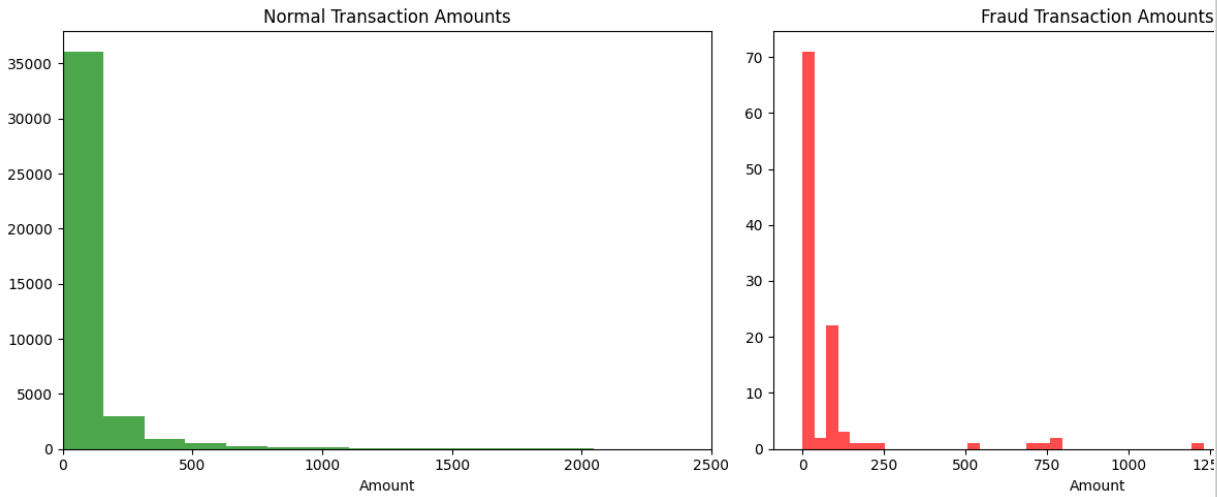




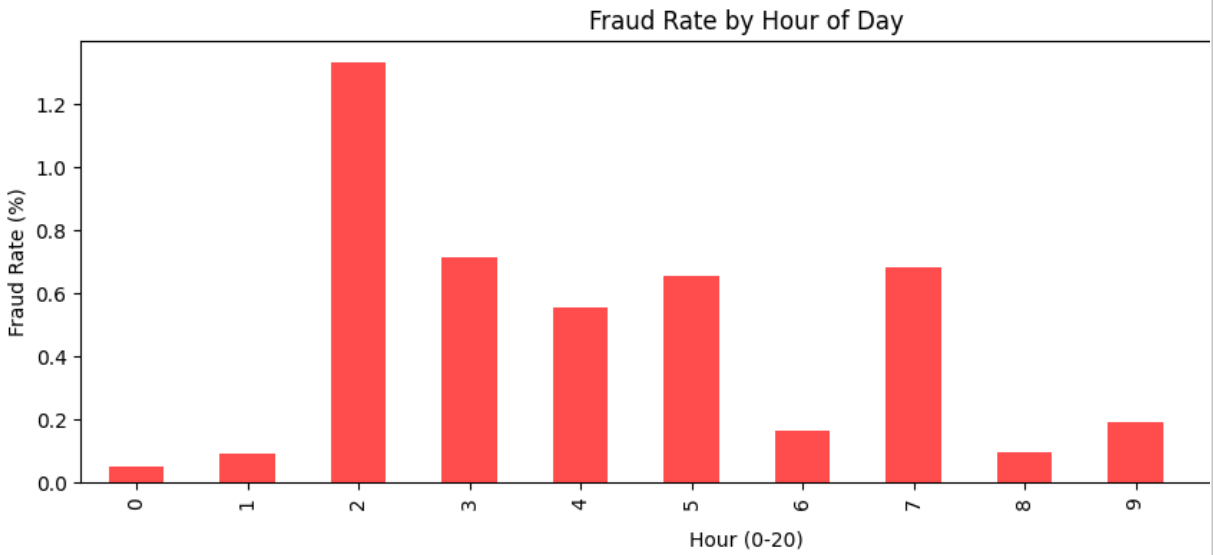


Bussiness Insights

```
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Normal transactions
axes[0].hist(df[df['Class']==0]['Amount'],
             bins=50, color='green', alpha=0.7)
axes[0].set_title('Normal Transaction Amounts')
axes[0].set_xlabel('Amount')
axes[0].set_xlim(0, 2500)

# Fraud transactions
axes[1].hist(df[df['Class']==1]['Amount'],
             bins=50, color='red', alpha=0.7)
axes[1].set_title('Fraud Transaction Amounts')
axes[1].set_xlabel('Amount')

plt.tight_layout()
plt.show()
```



```
df['Hour'] = (df['Time'] // 3600).astype(int)

fraud_by_hour = df.groupby('Hour')['Class'].mean() * 100

plt.figure(figsize=(12, 4))
fraud_by_hour.plot(kind='bar', color='red', alpha=0.7)
plt.title('Fraud Rate by Hour of Day')
plt.xlabel('Hour (0-20)')
plt.ylabel('Fraud Rate (%)')
plt.show()

print("Highest fraud hour:",
      fraud_by_hour.idxmax())
print("Lowest fraud hour:",
      fraud_by_hour.idxmin())
```




```
Highest fraud hour: 2
Lowest fraud hour: 10
```

```
# Sahi formula:
df['Hour'] = (df['Time'] // 3600).astype(int)

print(df['Hour'].unique())
print(df['Hour'].value_counts())
```

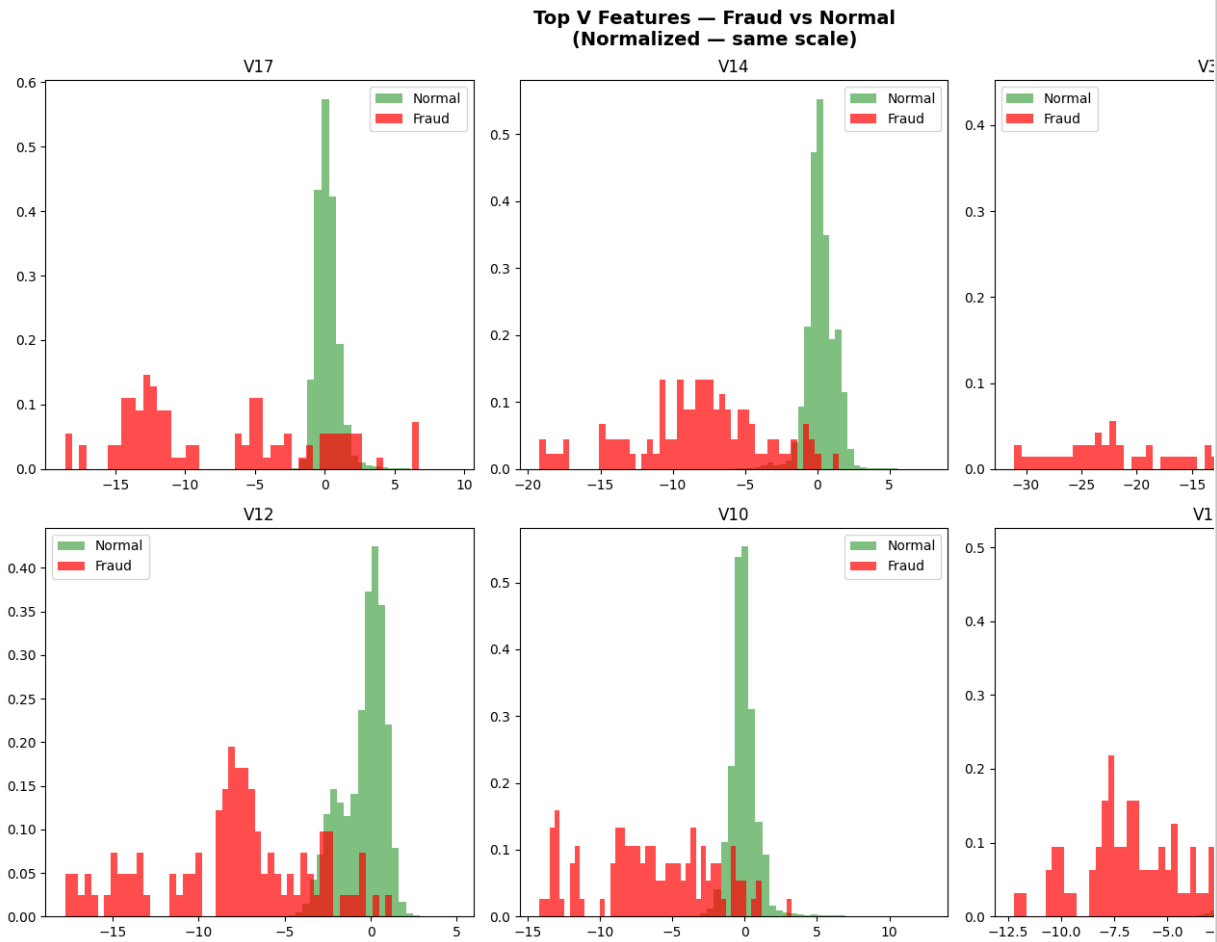
```
[ 0  1  2  3  4  5  6  7  8  9 10 11]
Hour
10    8288
9     7878
8     5179
0     3963
7     3368
11    2798
1     2217
6     1831
3     1821
5     1681
2     1576
4     1082
Name: count, dtype: int64
```

```
correlation = df.corr()['Class'].drop('Class')
correlations = correlation.abs().sort_values(ascending=False)
print("Top 10 features correlated with fraud:")
print(correlations.head(10))
```

```
Top 10 features correlated with fraud:
V3    0.388
V14   0.386
V17   0.382
V7    0.299
V10   0.294
V12   0.287
V16   0.260
V11   0.226
V4    0.206
V1    0.199
Name: Class, dtype: float64
```

```
fig, axes = plt.subplots(2, 3, figsize=(15, 10))
axes = axes.flatten()

top_features = ['V17', 'V14', 'V3',
                'V12', 'V10', 'V16']

for i, feat in enumerate(top_features):
    # Normal
    axes[i].hist(
```

```
        df[df['Class']==0][feat],
        bins=50, alpha=0.5,
        color='green', label='Normal',
        density=True  # ← Percentage mein
    )
    # Fraud
    axes[i].hist(
        df[df['Class']==1][feat],
        bins=50, alpha=0.7,
        color='red', label='Fraud',
        density=True  # ← Percentage mein
    )
    axes[i].set_title(f'{feat}')
    axes[i].legend()

plt.suptitle(
    'Top V Features — Fraud vs Normal\n'
    '(Normalized — same scale)',
    fontsize=14, fontweight='bold'
)
plt.tight_layout()
plt.show()
```



Feature Engineering

```
df['Amount_log'] = np.log1p(df['Amount'])

print("Before Transform")
print(f"Skewness: {df['Amount'].skew():.2f}")
print("\nAfter transform:")
print(f"Skewness: {df['Amount_log'].skew():.2f}")
```

```
Before Transform
Skewness: 10.13

After transform:
Skewness: 0.11
```

```
df['Is_night'] = df['Hour'].apply(lambda x : 1 if (x <= 6 or x>=20)
```

```
print("\nIs_Night fraud rate:")
print(df.groupby('Is_night')['Class'].mean()*100)
```

```
Is_Night fraud rate:
Is_night
0   0.182
1   0.409
Name: Class, dtype: float64
```

```
df['Amount_Category'] = pd.cut(df['Amount'],bins=[0,10,50,200,1000,
print("\nFraud rate by Amount Category:")
print(df.groupby('Amount_Category', observed=True)['Class'].mean()*
```

```
Fraud rate by Amount Category:
Amount_Category
Micro        0.454
Small        0.064
Medium       0.286
Large        0.162
Very Large   0.460
Name: Class, dtype: float64
```

```
df['V17_V14_Combined'] = df['V17'] * df['V14']
df['V3_V10_Combined']  = df['V3']  * df['V10']

# Verify
print("V17_V14_Combined correlation:")
print(df['V17_V14_Combined'].corr(df['Class']))

print("\nV3_V10_Combined correlation:")
print(df['V3_V10_Combined'].corr(df['Class']))
```

```
V17_V14_Combined correlation:
0.5379609379345955

V3_V10_Combined correlation:
0.5977334161302289
```

```
drop_cols = ['Amount','Time']
df = df.drop(drop_cols, axis=1)
df = pd.get_dummies(df, columns=['Amount_Category'], drop_first=Tru
print(f"Final shape: {df.shape}")
print(f"Columns: {df.columns.tolist()}")
```

```
Final shape: (41682, 38)
Columns: ['V1', 'V2', 'V3', 'V4', 'V5', 'V6', 'V7', 'V8', 'V9', 'V10
```

```
df.to_csv('fraud_features.csv', index=False)
print("✅ Saved!")
```

```
✅ Saved!
```

```
print(df.shape)
print(df.duplicated().sum())
```

```
(41682, 38)
701
```

```
df = df.drop_duplicates()
print(f"After dedup: {df.shape}")
```

```
After dedup: (40981, 38)
```

```
# Class distribution dekho
print(df['Class'].value_counts())
print(f"Fraud rate: {df['Class'].mean()*100:.3f}%")

# Original tha:
# Normal: 116,697
# Fraud : 244
```

```
Class
0    40873
1      108
Name: count, dtype: int64
Fraud rate: 0.264%
```

```
# ==============================================
# FRAUD DETECTION — Fresh Start
# ==============================================
import pandas as pd
import numpy as np
import warnings
warnings.filterwarnings('ignore')
```

```
# Load
df = pd.read_csv('creditcard.csv')
print(f"1. Loaded: {df.shape}")

# Clean
df = df.dropna()
df['Class'] = df['Class'].astype(int)
print(f"2. Cleaned: {df.shape}")

# Features
df['Hour'] = (df['Time'] // 3600).astype(int)
df['Amount_Log'] = np.log1p(df['Amount'])
df['Is_Night'] = df['Hour'].apply(
    lambda x: 1 if (x <= 6 or x >= 20) else 0)
df['Amount_Category'] = pd.cut(
    df['Amount'],
    bins=[0, 10, 50, 200, 1000, 20000],
    labels=['Micro','Small','Medium',
            'Large','Very_Large']
)
df['V17_V14_Combined'] = df['V17'] * df['V14']
df['V3_V10_Combined']  = df['V3']  * df['V10']

# Drop
df = df.drop(columns=['Amount', 'Time'])

# Dummies
df = pd.get_dummies(
    df, columns=['Amount_Category'],
    drop_first=True
)

print(f"3. Final: {df.shape}")
print(f"4. Fraud rate: {df['Class'].mean()*100:.3f}%")
print(f"5. Duplicates: {df.duplicated().sum()}")
```

```
1. Loaded: (41683, 31)
2. Cleaned: (41682, 31)
3. Final: (41682, 38)
4. Fraud rate: 0.259%
5. Duplicates: 701
```

```
# Duplicates remove karo
df = df.drop_duplicates()
print(f"After dedup: {df.shape}")
print(f"Fraud cases: {df['Class'].sum()}")
print(f"Fraud rate : {df['Class'].mean()*100:.3f}%")
```

```
After dedup: (40981, 38)
Fraud cases: 108
Fraud rate : 0.264%
```